

Algebra First-Year Exam, August 2020, Part 1

Answer four problems. Write your answers clearly in complete English sentences.

1. Denote by Q_8 the quaternion group. Show that if $f : Q_8 \rightarrow S_n$ is an injective homomorphism then $n \geq 8$.
2. Let G be a group.
 - (i) Show that if G is abelian, the subset H of elements of finite order is a subgroup.
 - (ii) Give an example of a (nonabelian) group G and elements of finite order $x, y \in G$ such that xy has infinite order, and prove that this example is correct.
3. Prove the first assertion of Sylow's theorem: if G is a group of order n , p is prime and p^a is the largest power of p dividing n , then G has a subgroup of order p^a . State clearly any preliminary results you need.
4. This problem has three parts.
 - (i) Define what it means for a group G to be nilpotent.
 - (ii) Show that any nilpotent group is solvable.
 - (iii) Give an example of a finite solvable group that is not nilpotent, and prove that this example is correct.
5. Let G be a group acting on a set S , and $H \subseteq G$ a normal subgroup. Assume that for any $x_1, x_2 \in S$ there is a unique $h \in H$ such that $h(x_1) = x_2$. For $x \in S$ let G_x be the stabilizer of x . Show that for any $x \in S$, G is a semi-direct product of G_x and H . Do not assume that G is finite.
6. In both cases, justify your answer.
 - (1) Find all conjugacy classes of elements of order 2 in S_6 .
 - (2) Do the same for A_6 .
7. Show that the normalizer of a 5-Sylow subgroup of A_5 has order 10, and is a maximal subgroup of A_5 .